IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 11 August 2024, accepted 2 September 2024, date of publication 11 September 2024, date of current version 22 October 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3457866

RESEARCH ARTICLE

Beyond the Illusion: Ensemble Learning for Effective Voice Deepfake Detection

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This work was supported in part by the University of Okara; and in part by the Department of CS and SE, International Islamic University, Islamabad, Pakistan.

ABSTRACT Deepfake synthetic media, manipulated using artificial intelligence to mimic authenticity, has become more dangerous in the modern digital era. Despite significant progress in video deepfake detection, audio deepfake detection relies on specific datasets and machine learning algorithms. This study addresses this limitation by developing a deep ensemble learning approach. The proposed deep ensemble approach for audio deepfake detection is called the Ensemble Convolutional Neural Network-Mel Frequency Cepstral Coefficient (ECN-MF). The ECN-MF model comprises a recurrent neural network, a 1D convolutional neural network, long short-term memory networks, and a convolutional long short-term memory network to extract a wider range of audio features, including mel-frequency cepstral coefficient, chroma features, and zero crossing rate. The data processing step was carried out through the creation of a preprocessing pipeline, inclusive of feature extraction, dimensionality standardization, and data normalization. The summarized features were incorporated into the feature vector, which was then normalized before being standardized in order to enhance consistency and stability across the audios. The investigation of the suggested ECN-MF model was carried out using the Fake-or-Real dataset. The dataset comprises four sub-datasets ('for-original', 'for-norm', 'for-2sec', and 'for-rerec'), categorized by audio duration and bit rate. To evaluate the performance of the proposed ensemble model, we utilized all sub-datasets with fake and real audio. The proposed ensemble approach achieved state-of-the-art accuracies of 99.5% on the 'for-original' sub-dataset, closely matching the CNN model at 99.6%. It also achieved accuracies of 98% on 'for-norm,' 96.9% on 'for-2sec,' and 92.8% on 'for-rerec' sub-datasets. By applying the proposed ensemble model to the 'for-merged' dataset, which comprises all sub-datasets, we obtained a state-of-the-art accuracy of 98%. These results demonstrate the effectiveness of the proposed approach, which outperforms the results of the individual models.

INDEX TERMS Voice deepfake, fake-or-real dataset, ensemble model, RNN, 1D CNN, LSTM, ConvLSTM, machine learning approach.

I. INTRODUCTION

The exponential growth of deepfake technology represents one of the greatest challenges in the digital world. Deepfakes, deliberately modified or synthesized media content created

The associate editor coordinating the review of this manuscript and approving it for publication was Mostafa M. Fouda.

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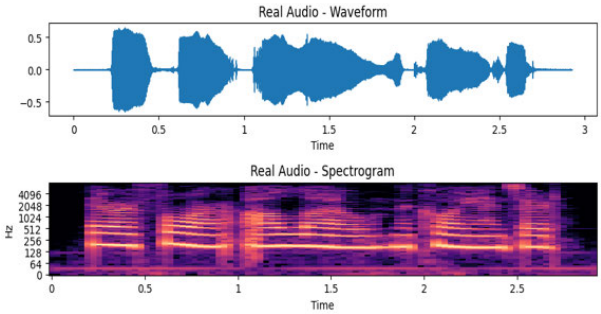

FIGURE 1. Waveform and spectrogram of real audio signal.

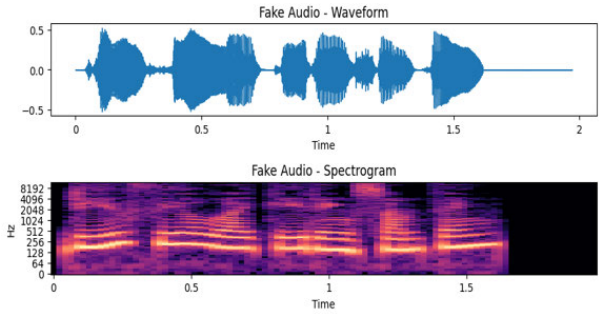

FIGURE 2. Waveform and spectrogram of fake audio signal.

using advanced machine-learning algorithms [1], were first termed "Deepfake" in late 2017 by a Reddit user of the same name. This user created a space on the online news and aggregation site where they shared pornographic videos utilizing open-source face-swapping technology [2]. At this early stage of technological development, it inevitably introduced a host of legal issues and raised societal concerns. The potential risks of deepfakes are significant, as they can be used to create false evidence, spread fake news, or commit financial fraud and other crimes [3].

Applying machine learning models is common across various fields, such as ranking [4] and automating language recognition [5]. Recent technological developments and the emergence of social media have given these applications new life. Deepfake audio refers to manipulating audio recordings to produce fake speech and related information [6]. The rapid advancement of Text-To-Speech (TTS) synthesis and Voice Conversion (VC) techniques has notably increased the possibility of impersonating a person's voice [7]. Studies have shown that in the worst cases, deepfake audio has been used to alter crucial evidence, potentially allowing criminals to evade justice [8]. Therefore, the ability to detect and recognize these fraudulent audio recordings has become of paramount importance. Identifying deepfakes across various mediums, including audio, video, and text, is a highly explored and actively developing research domain. While recent works have predominantly focused on deepfake video detection, achieving greater accuracy [9], deepfake audio detection is still in its early stages compared to methods employed for images and videos.

In these areas, a combination of audio and spatiotemporal data is frequently utilized to train robust deep-learning models. Despite this relative lag in audio-focused advancements, the importance of audio-based classifiers in classification and detection cannot be overstated. The visual content of the waveform and spectrogram of the real and fake audio signals are illustrated in Figures 1 and 2, where the waveform displays the temporal features, and the spectrogram captures the frequency-related patterns.

This study explores the efficacy of ensemble learning techniques (soft voting) in increasing the power of deep learning. The main purpose of the ensemble approach is the fusion of several deep learning models to improve overall performance. The core of ensemble learning is combining the advantages of many deep learning algorithms into a single system. This ensemble approach uses prominent models such as Recurrent Neural Network (RNN), 1-dimensional Convolutional Neural Network (1D CNN), Long Short-Term Memory (LSTM), and Convolutional LSTM (ConvLSTM) as the essential parts, and together, they improve the accuracy of audio classification. We used the Fake-or-Real dataset for the experimentation with great precision. This dataset is an extensive collection of four different sub-datasets. It was suggested that the ensemble technique guaranteed great assessment by using all sub-datasets. It clearly showed better performance compared to the individual base models. This thorough study shows the significant power of ensemble learning in addressing the challenges of identifying altered audio information.

Also, the ASVSpoof dataset has played a pivotal role in advancing the accuracy of automatic speaker verification systems. The ASVspoof2015 dataset [10] greatly developed automatic speaker verification spoofing and countermeasures, reducing the equal error rate (EER) by nearly 1.5%. However, specific attacks still exhibited EERs as high as 50%, and unknown attacks returned EERs five times greater. Subsequent iterations, such as ASVspoof2017 [11], further improved the detection of replay spoofing attacks and achieved a lowered EER of 6.73%. In ASVspoof2019 [12], a more pronounced emphasis was placed on developing countermeasures for automatic speaker verification and identifying spoofed audio. ASVspoof2021 [13] also evaluated speech processing in real-world scenarios and highlighted challenges in deepfake detection, replay attack detection, and compression resilience.

Furthermore, feature extraction techniques are used to improve the precision and robustness of the detection system. Three key features have been identified and extracted from the audio, namely Mel-Frequency Cepstral Coefficients (MFCC), Chroma, and Zero Crossing Rate [14], [15], [16]. The importance of using these features in audio analysis has been carefully considered. The extraction of these features offers a comprehensive understanding of the dataset. Their incorporation with the suggested ensemble approach

improved the validity of the outcomes by allowing the investigation of a wider range of distinct qualities.

- Introduction of an ensemble learning-based method for deepfake detection.
- Utilization of the sub-datasets of the Fake-or-Real dataset for experimentation.
- Creation of the 'for-merged' dataset, which involved combining all four sub-datasets.
- Application of a feature extraction methodology to extract MFCC, Chroma, and ZCR features from the audio sources.
- Demonstration of the superior performance of our proposed ensemble model compared to other deep learning models.

The remainder of this paper is structured as follows: Section II extensively reviews the literature on methods for detecting audio deepfakes. Section III offers a comprehensive exposition of our proposed methodology. The subsequent section, Section IV, comprehensively addresses results and discussion, while Section V serves as a conclusion and outlines potential avenues for future research. This streamlined structure ensures a thorough exploration of the topic and provides a clear roadmap for understanding the study's findings and guiding future directions.

II. LITERATURE REVIEW

Significant attention has been paid to deepfake audio detection in recent years due to the combined efforts of researchers and industry experts. They aimed to minimize the potential risks and the negative impact of deepfake technology. Many approaches and methods have been developed to address the challenges of identifying fake audio material. Additionally, there was a rise in interest in applying deep learning to detect deepfake audio in the latter half of the 2010s. Deep learning models provided advantages in the recognition process by understanding the data and helped improve the detection system's accuracy. This section offers a comprehensive assessment of previous studies focused on examining contemporary methods and updates in deepfake audio detection that involve deep learning models.

Many studies used the ASVspoof Challenge 2019 dataset for audio deepfake detection. The study [17] implemented the deep neural network architecture based on SENet and combined F0 information with real and imaginary spectrogram features extracted from the speech waveform. This approach achieved an equal error rate (EER) of 0.43%, showing the potential for audio deepfake detection. The Breathing-Talking-Silence Encoding (BTS-E) architecture was proposed and divided into two training stages: Deepfake Detection and Self-segmentation [18]. The BTS-E system exhibited 2x performance enhancements in different types of attacks compared to similar works while reacting to ASVspoof2019 and ASVspoof2021 datasets, resulting in 2x to 7.76% average EER and t-DCF reduction. The researchers [19] introduced the development of the detection of fake without forgetting (DFWF) purpose, a learning strategy of isolating and recognizing fake content. DFWF produced relative reductions in AvgEER of 19.08% for "PA → LA" and 30.29% for "LA → PA". A score fusion-based method produced scores for each audio sample, integrating AM-Softmax Loss and Center Loss [20]. For the first and second rounds of the ASVspoof 2019 LA dataset, the study obtained a weighted equal error rate (WEER) of 10.93% and 11.96%, respectively. A research study proposed the RawGAT-ST model, which used graph fusion techniques and graph attention networks (GAT) to achieve an EER of 1.06% [21]. Furthermore, [16] described a gamified classification challenge in which an AI algorithm competed against human players in audio deepfake detection. For the AI algorithm, the accuracy was 95%, which outperformed the humans, who achieved 80% accuracy. These studies collectively demonstrated the potential of using the ASVspoof 2019 dataset and various techniques in detecting manipulated voice content.

The studies [22], [23] used the wav2vec feature extraction technique to detect deepfake audios. The study [23] developed a wav2vec2 model. It was a pre-trained feature extractor combined with a downstream classifier for deepfake detection. The wav2vec2 model extracted features from the audio data, which were then fed into the classifier for classification purposes. The ADD challenge dataset and the ASVspoof 2021 challenge dataset were used for this purpose. The proposed system showed an EER (%) of 3.54 on LA and 4.98 on DF. Reference [22] introduced a three-module strategy. wav2vec features, phoneme duration features, and pronunciation features are the three categories of features that may be extracted by the first module, which is a feature extractor. The second module combined wav2vec data with prosodic and pronunciation features of varying weights to create an attention module. The third module, which learned a deep representation of speech, was a back-end. It used the ASVspoof2019, ASVspoof2015, and VCC2020 In-the-Wild ADD2022 databases. On the ASVspoof2015 test set, the suggested approach produced an equal error rate (EER) of 3.96%, a notable improvement over the EER of 6.59% obtained using simply the wav2vec features. Reference [24] presented a novel deepfake detection system that used cross-modal learning, optical flow, ensemble learning, and audio-video-textual characteristics. The authors utilized the DfTIMIT, DFDC, and CustomDataset datasets. In tests on CustomDataset, DFDC, and DfTIMIT, this system performed at 87.6%, 74.2%, and 93.2%, respectively. Reference [9] proposed a feature extraction approach using Mel-frequency Cepstral Coefficients (MFCC) from audio sources. They then used SVM, DT, RF, LSTM, and deep learning models, such as VGG16 and VGG19, to classify the audio as real or fake.

Some studies introduced the CNN model with other deep-learning techniques. Lim et al. [25] used explainable artificial intelligence (XAI) techniques for detecting deepfake voices. The study focused on multi-modal interpretability at a human

perception level, using visual and audio interpretation to enable intuitive recognition of characteristics. Additionally, the authors used a simple convolutional neural network and LSTM for deepfake voice detection and a spectrogram for feature extraction from the raw audio data. CNN and CNN-LSTM showed promising results. Yi et al. [26] described a model that includes benchmarking experiments using the Gaussian mixture model (GMM) and light convolutional neural network (LCNN) to train baseline models. It also described the feature extraction process using constant Q cepstral coefficients (CQCC) and the evaluation metric used to assess the performance of the models. For the partially fake test set, the GMM model achieved 43.17%, 87.52%, and 56.37 and the LCNN model gained 71.62%, 93.86%, and 82.15% precision, recall, and F1 scores, respectively. The study [27] introduced the use of a Convolutional Neural Network (CNN) and a Support Vector Machine (SVM) for the identification of fake stereo audio. The CNN extracts features from the audio signals, while the SVM was used for classification. The accuracy rate for the intra-dataset evaluation was 99.5%, and the accuracy rate for the cross-dataset assessment was 99%. Ustubioglu et al. [28] used deep learning with Mel spectrogram to detect audio copy-move forgery. Specifically, a Convolutional Neural Network (CNN) architecture was proposed to classify suspicious Mel spectrograms. The accuracy and recall values of the proposed method were above 0.98, and the precision and F1-score values were above 0.95 and 0.97, respectively. The research [7] proposed a model composed of two blocks: a Speech Emotion Recognition (SER) system and a Synthetic Speech Detector (SSD) system. The SER system extracts features from an input speech signal using a 3D convolutional Recurrent Neural Network (CRNN). The SSD system then associates a class with the input features using a Random Forest (RF) classifier. The studies [29], [30], [31] proposed different versions of the ResNet model. Yan et al. [29] introduced a ResNet-18 model that utilizes two stacked convolutional layers with 3×3 kernels per building block, subsampling with a stride of 2, followed by global average pooling and a fully connected layer, ultimately providing classification results for eight vocoders. They used a vocoder fingerprint detection dataset; all of the audio samples in this dataset are from Chinese AISHELL3. The proposed method achieved high accuracy in detecting vocoder fingerprints using LFCC features and a ResNet model, with F1-score values ranging from 96.68% to 99.99%. Reference [30] introduced a Residual Network (ResNet-34) model with multi-head attention pooling and neural stitching for deepfake audio detection. The authors used the datasets provided by the Anti-Deepfake Challenge 2022 (ADD 2022). Their system achieved first place in Track 3.2 with an official EER of 8.6% and 11.1% for the 1st and 2nd rounds, respectively. Reference [31] developed a model architecture that included components such as ResNet-1D, CNN-1D, Transformer encoder, BiLSTM, and fully connected layers for waveform boundary detection of partially spoofed audio. This system achieved an equal error rate (EER) of 6.58% on the ADD2022 challenge test set.

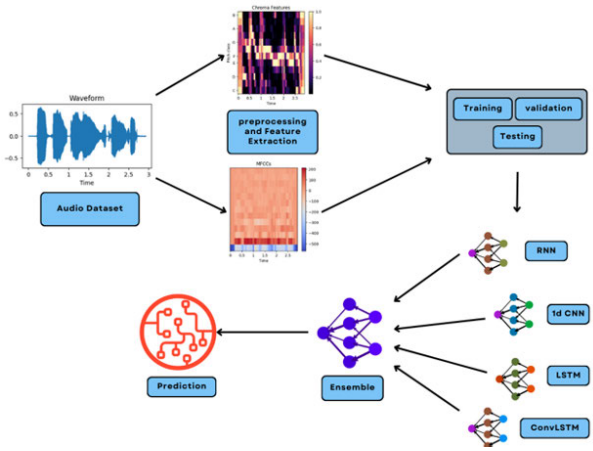

FIGURE 3. Block diagram of the proposed model.

Improving machine learning model performance necessitates incorporating feature-based techniques. Deep learning methods, on the other hand, produce superior results. Consequently, this research explores the capabilities of deep learning approaches for identifying deepfake content.

III. MATERIALS AND METHODS

This study endeavors to develop a technique for classifying deepfake synthetic audio, considering various factors such as audio sizes and durations. The approach introduced here presents a comprehensive framework designed to handle extensive training datasets and employ ensemble learning with deep learning algorithms for detection. The proposed framework will be explained in the subsequent section, encompassing sub-datasets, preprocessing steps, feature extraction, and the suggested models. Figure 3 provides a detailed architectural depiction of the framework, consisting of four core components: 1) Dataset, 2) Preprocessing and Feature Extraction, and 3) Proposed Ensemble Approach. A detailed description of each phase follows:

A. DATASET

Fake-or-Real (FoR) [32] consists of 195,000 audio samples that were generated using state-of-the-art speech synthesis techniques implemented by a computer. It is a composite dataset that combines various datasets from different investigations. The primary purpose of creating this dataset was to train algorithms in identifying counterfeit speech [33]. The FoR dataset comprises four subsets, namely: (i) for-orig or for-original, (ii) for-norm, (iii) for-2sec, and (iv) for-rerec. The trials carried out for this study made use of all four subsets. The 'for-orig' dataset preserves its original properties and is not modified from its source. To maintain gender and class equity, the 'for-norm' subset duplicates the 'for-orig'

TABLE 1. Dataset details.

Dataset	Classes	Samples	Total
for-orig	Real	34,644	**69,374**
	Fake	34,730	
for-norm	Real	30,294	**65,049**
	Fake	34,755	
for-2sec	Real	8,945	**17,890**
	Fake	8,945	
for-rerec	Real	6,613	**13,268**
	Fake	6,655	
for-merged	Real	80,496	**165,581**
	Fake	85,085	

while standardizing the sampling rate, size, and bandwidth. The 'for-2sec' subset consists of audio recordings truncated to a fixed two seconds. On the other hand, the 'for-rerec' subset adds variation to the dataset by simulating audio sent by an attacker and including duplicates, 0-bit files, and different bit rates.

An important decision was made to combine individual subsets into a combined dataset called the 'for-merged' dataset for experimentation. Creating a single enriched dataset aimed to build a more reliable and broadly applicable model. This move greatly increases the performance of the ensemble model. This dataset included a variety of samples to improve model accuracy and precision. Table 1 provides full information on all subsets, including the number of samples used for each dataset.

B. DATASET SPLITTING

Machine learning model creation involves separating datasets into three parts: evaluation, training, and validation. Ensuring success through this method provides many substantial advantages. First, it offers a glimpse into the model's performance using many sample data sets; hence, it is easy to evaluate whether or not the model can generalize to new (unknown) data. This kind of separation helps prevent overfitting, one of the common errors when a model starts to understand the training data too much and, in turn, produces poor estimates. Another aspect is that the tuning of hyperparameters highly depends on the validation set, which makes it possible to set the model's settings in preference to an increase in performance without affecting the testing set. With that being said, partitioning, of course, makes the whole process comparable, which can be utilized to show how each method or model performs. To guarantee that predicting models are precise, reliable, and operable in scenarios, it is imperative to classify the datasets into different types of groupings wisely. This branch addresses the creation, shaping, and validation of the models.

The dataset described is organized into five main categories: for-orig, for-norm, for-2sec, for-rerec, and for-merged. Each category breaks down the data into training and validation sets and then tests it on the one-off-the-shelf set. Each partition has examples of data labeled as either "Real" or "Fake," and the number of examples determines hybrids proportionately. Table 2 describes the details of the dataset splitting of all datasets used in experiments.

TABLE 2. Dataset splitting.

Dataset	Split	Classes	Samples	Sub Total Samples	Total
for-orig	Training	Real	26971	53903	**69374**
		Fake	26932		
	Validation	Real	2274	4664	
		Fake	2390		
	Testing	Real	5399	10807	
		Fake	5408		
for-norm	Training	Real	22630	49617	**65049**
		Fake	26987		
	Validation	Real	2264	4634	
		Fake	2370		
	Testing	Real	5400	10798	
		Fake	5398		
for-2sec	Training	Real	6988	13976	**17890**
		Fake	6988		
	Validation	Real	544	1088	
		Fake	544		
	Testing	Real	1413	2826	
		Fake	1413		
for-rerec	Training	Real	5104	10208	**13268**
		Fake	5104		
	Validation	Real	408	816	
		Fake	408		
	Testing	Real	1101	2244	
		Fake	1143		
for-merged	Training	Real	61693	127704	**165581**
		Fake	66011		
	Validation	Real	5490	11202	
		Fake	5712		
	Testing	Real	13313	26675	
		Fake	13362		

C. PRE-PROCESSING AND FEATURE EXTRACTION

The file paths for the training, testing, and validation datasets set up the pipeline for data preprocessing, laying the foundation for efficient data handling in later processing phases. We extracted key features such as chroma features, Mel-frequency cepstral coefficients (MFCCs), and zero-crossing rates (ZCR) from the audio content. These features were combined to create a single feature vector, providing a comprehensive depiction of the audio signals. Below is a summary of the key details of the extracted features.

1) MEL-FREQUENCY CEPSTRAL COEFFICIENTS (MFCC)

MFCC is a commonly used technique in speech and audio processing that approximates the responses of the human auditory system, offering a clear representation of the spectral shape of an audio signal. This technique helps distinguish between different sound-generating methods by capturing the temporal characteristics of sound. We focused on the average value of 13 MFCC coefficients for analysis. The calculation of MFCC involves multiple steps: computing the Mel spectrogram, applying the Discrete Cosine Transform (DCT), performing the short-time Fourier Transform (STFT) on each

frame, and selecting specific coefficients. The equation is:

$$\mathrm{MFCC}(n) = \mathrm{DCT}\left(\log\left(\sum_{m=0}^{M-1} \mathrm{MelFilter}(m,f)\cdot|\mathrm{STFT}(n,f)|^2\right)\right) \tag{1}$$

2) CHROMA

Chroma is essential in audio analysis and music. Chroma features highlight the pitch content and pitch class distribution. They are useful for chord detection and music categorization. The equation for Chroma is as follows:

$$C_f(b) = \sum_{z=0}^{Z-1} \left|X_{lf}(b+z\beta)\right| \tag{2}$$

3) ZERO CROSSING RATE (ZCR)

ZCR monitors variations in the signal to determine the smoothness or noisiness. It is useful for several types of audio processing and helps identify sudden changes in sound. ZCR for a signal s of length T can be defined as:

$$ZCR = \frac{1}{T}\sum_{t=1}^{T} |s(t) - s(t-1)| \tag{3}$$

Implementing these feature extraction approaches extracted valuable information from the audio data, making the analysis process more economical and effective for our particular goals.

4) STANDARDIZING DIMENSIONS

Standardizing dimensions was applied to the feature vector. This was a crucial stage in the preprocessing workflow. We achieved uniformity and consistency in each audio sample by truncating or padding based on the minimal number of frames (*min_frames*) across all features. It is mathematically expressed as:

$$min_{frames} = \min(ZCR_{frames}, MFCCs_{frames}, Chroma_{frames}, max_{frames}) \tag{4}$$

$$PaddedFeature = np.concatenate((Feature, padding), axis = 1) \tag{5}$$

The feature vectors were then truncated or padded accordingly:

$$TruncatedFeature = Feature[:, min_{frames}] \tag{6}$$

5) NORMALIZATION

Normalization improves the stability of our data. It ensures that the data from various features have a consistent scale. The feature vectors were normalized using a standard scalar. This normalization process is expressed as:

$$NormalizedFeature = \frac{Feature - Feature.mean()}{Feature.std()} \tag{7}$$

```
Algorithm: Ensemble Model

Step 1. Load Pre-trained Models: load_model(cnn_model, lstm_model, rnn_model, cnn_lstm_model)
Step 2. Define Ensemble Model Function:
  function ensemble_soft_voting(models, input_layers):
    model(input_layer) for model, input_layer in zip(models, input_layers)]
    Average(outputs)
    Model(inputs=input_layers, outputs=avg, name='ensemble_soft_voting')
    return model
Step 3. Create Input Layers: input_layers = [Input() for _ in range(4)]
Step 4. Create Ensemble Model: ensemble_model = ensemble_soft_voting([rnn, cnn, lstm, cnn_lstm],
                                input_layers)
Step 5. Compile and Evaluate Ensemble Model: ensemble_model.compile(), ensemble_model.fit(),
                                ensemble_model.evaluate()
Step 6. End of Algorithm
```

FIGURE 4. Ensemble model algorithm.

These techniques prevent any single feature from having an outsized influence. We also enhance the learning process and facilitate effective pattern recognition by the models.

We chose the dataset carefully by integrating different preprocessing methods, such as dimension standardization, normalization, and feature extraction. As with transistor representations and normalized features, these heavy and persistent changes to the data provide a precedent for building and testing machine learning models. The resultant models proved to be very accurate and had a wide generalization aptitude, and they can overcome the complexities of many types of audio data.

D. THE PROPOSED ENSEMBLE MODEL

To conduct the experiments, we implemented an ensemble learning technique (soft voting) that prioritizes outcomes. Four different base models are used in the ensemble model, which is carefully designed to perform best in binary classification tasks: Recurrent Neural Network (RNN), Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), and Convolutional Long Short-Term Memory (ConvLSTM). The ensemble generates a collective output based on the soft voting concept by averaging the predictions from each base model.

The Adam optimizer is then used to construct our model, employing a learning rate schedule and binary cross-entropy as the loss function. With a significant 410,692 trainable parameters, our ensemble model has an enhanced ability to learn and adapt during training. This high number of parameters helps the model better capture complex patterns and subtleties in the data, which leads to improved learning and performance results. Figure 4 shows the algorithm of the ensemble model, and Figure 5 shows the ensemble model architecture.

An early stopping mechanism is included to reduce the risk of overfitting that may arise during the training of our ensemble model. This technique avoids unnecessary training that might impair the model's generalization ability by routinely tracking validation loss throughout epochs and acting when overfitting is suspected.

Combining the predictions using the average layer from the separate outputs of the base models yields the ensemble model's final output. These foundational models have unique characteristics and architectural elements specially adapted

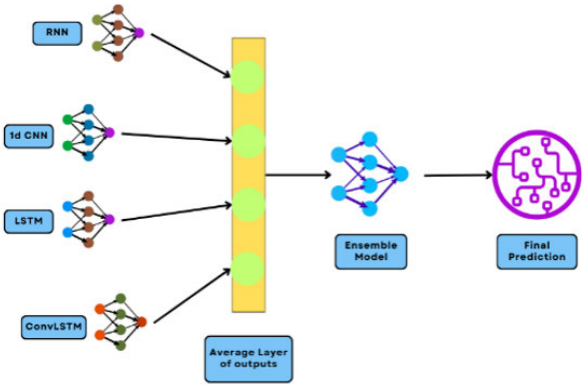

FIGURE 5. The proposed ensemble model architecture.

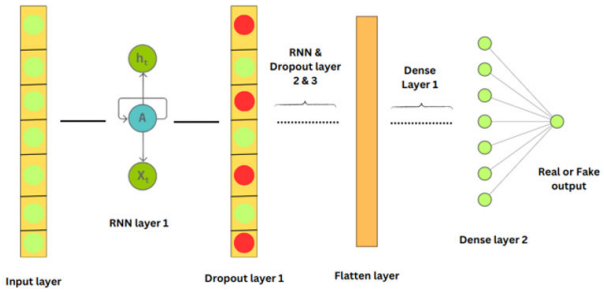

FIGURE 6. RNN architecture.

for binary classification tasks. They are built as sequential neural networks. These models are important building blocks of the proposed ensemble model, which leads to higher accuracy in the audio classification domain.

Figure 5 shows details of each particular base model to ensure full understanding. Comprehensive understanding is generated by offering detailed information on the specifications of each model. Combining the models improves the accuracy and precision of the ensemble approach in determining binary classification outcomes.

The proposed ensemble model offers several advantages over state-of-the-art Graph Neural Networks (GNNs). Firstly, the ensemble approach combines the strengths of multiple base models, such as RNN, 1D CNN, LSTM, and ConvLSTM, with improved performance and robustness. This hybrid structure allows the model to capture a wider range of patterns and features in the data. It enhances the model's ability to generalize across different datasets and conditions. Additionally, the ensemble method mitigates the weaknesses of individual models, thereby reducing the risk of overfitting and improving overall accuracy. Empirical results demonstrate that the ensemble model achieves higher accuracy rates than the GNNs, especially in sub-datasets. This performance boost underscores the model's capability to handle the complexity and variability inherent in deepfake detection tasks. Moreover, the theoretical analysis supports the ensemble approach, validates its empirical success, and highlights its superior ability to tackle the nuanced challenges posed by synthetic data.

1) RECURRENT NEURAL NETWORK (RNN)

Figure 6 shows the integration of the Recurrent Neural Network (RNN) model. This model features a sequential architecture comprising multiple Simple RNN layers. In this configuration, the structure progresses step by step: a first Simple RNN layer containing 64 units integrates smoothly with a Dropout layer, which serves for regularization. This Dropout layer helps to avoid overfitting throughout the model's training phase. The second Simple RNN layer consists of 128 units, double that of the first. Another Dropout layer reduces overfitting and improves the model's generalization. Then, the third Simple RNN layer reduces the output to 64 units and combines it with an extra Dropout layer for regularization.

The model is completed when a Flattened layer is added, and the output from the preceding layers is reshaped. This reshaping prepares the data for further processing. Subsequently, two Dense layers are incorporated. The first layer comprises 32 units, and the second contains one unit. These Dense layers contribute to achieving the final RNN architecture, enhancing its ability to capture complex patterns and features.

a: RATIONALE FOR CHOOSING RNN

RNNs are good for sequential data, thus making them ideal for audio. Unlike image signals, audio signals are, by default, time series that contain temporal relations, which is important when discriminating between real and fake audio. RNNs can pass their state information from one time step to another, which makes them useful in identifying such temporal patterns. This characteristic can help RNNs investigate the sequence of the audio frames, and they are quite useful for such applications as ASR and, in this case, detecting deepfakes.

2) 1D CONVOLUTIONAL NEURAL NETWORK (1D CNN)

The 1D CNN is a sequential architecture composed of multiple Conv1D (1D Convolutional Neural Network) layers, each designed to extract and process spatial hierarchies of features from the input data. The first Conv1D layer with 64 units extracts various characteristics from the input data. Then, the output is down-sampled using a MaxPooling1D layer to increase computing efficiency. The second Conv1D layer, applied with 128 units, provides more feature representation. This layer is added to the network to make the model more efficient. Then, the data is further condensed with another MaxPooling1D layer. A third Conv1D layer follows, which amplifies the feature extraction with a higher number of 128 units, succeeded by a fourth MaxPooling1D layer. An added Flatten layer is crucial for organizing the spatial information effectively, followed by Dense layers that prepare for processing the outputs of the convolutional layers. The first Dense layer consists of 128 units that enhance the

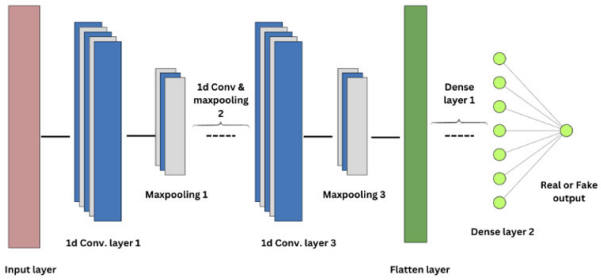

FIGURE 7. 1D CNN architecture.

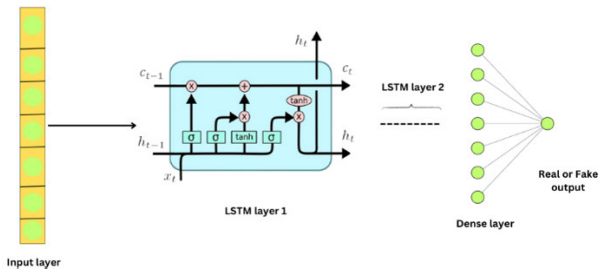

FIGURE 8. LSTM architecture.

pattern recognition capabilities of the model. Finally, the architectural design ends with a Dense layer containing a single unit to facilitate the specific 1D CNN architecture for audio categorization. The fusion of these Conv1D layers, MaxPooling1D layers, and Dense layers optimally configures the model for audio classification tasks. The architecture of the 1D CNN is depicted in Figure 7.

a: RATIONALE FOR CHOOSING 1D CNN

1D Convolutional Neural Networks (1D-CNNs) are extremely efficient in extracting local features along with MFCC, Chroma, and ZCR from the audio signals. Unlike 2D-CNNs that are commonly employed in image processing, 1D CNNs work with one-dimensional inputs and, therefore, are appropriate for audio waveform analysis. By employing convolutional filters along the time dimension of the audio signal, 1D-CNNs can detect features such as frequency transitions, pitch variations, etc., that are characteristic of deepfake audio. The incorporation of the local dependencies helps the overall extraction of the features in the model and provides detailed information on audio signals.

3) LONG SHORT-TERM MEMORY (LSTM)

The LSTM (Long Short-Term Memory) model is designed within a sequential architecture. It is specifically crafted to leverage the unique features inherent in the LSTM layers, making it highly effective for handling sequential data. The model consists of two LSTM layers, each containing 64 units. These layers help the model enhance its capacity to find patterns in the provided audio dataset. The final layer of the architecture is a single-unit Dense layer that employs a sigmoid activation function, making the model well-suited for binary classification tasks such as deepfake detection of audio content. Due to the simplicity and compact size of the LSTM model, it is particularly effective for applications with limited computational resources. It manages computational resources efficiently and handles sequential data adeptly due to its internal structure, which takes previous output as input for the next data point and addresses the vanishing gradient problem. Its streamlined design and effective resource utilization make it a preferred choice for tasks requiring accuracy and computational economy. An architectural overview of the LSTM is shown in Figure 8.

a: RATIONALE FOR CHOOSING LSTM

Long Short-Term Memory (LSTM) networks are a kind of RNN that mitigate the shortcomings of basic RNNs, especially the vanishing gradient problem. The fact that LSTMs have memory cells that allow data to be stored during long sequences enables them to learn long-term dependencies in audio signals. This capability is very important for audio deepfake detection because characteristics that differentiate fake and real audio may be dispersed over time. Due to their ability to retain information over longer sequences, LSTMs are better equipped to recognize the contextual format of audio data, thereby enhancing the performance of the proposed model.

4) CONVOLUTIONAL LSTM (CONVLSTM)

The proposed ConvLSTM model is designed by combining LSTM and Conv1D layers to leverage the combined effects of both. Finely tuned for binary classification problems, the first phase of this model involves extracting and down-sampling features hierarchically using three Conv1D layers, each cleverly linked with a MaxPooling1D layer. Following this, two LSTM layers are added to enhance the network's understanding of complex sequential patterns. The first LSTM layer skillfully returns sequences, while the second LSTM layer creates the final representation by combining the learned features. The model concludes with a Dense layer fitted with a sigmoid activation function to provide a final, binary classification-specific outcome. Figure 9 demonstrates the capabilities of the ConvLSTM model, making it a good fit for deepfake detection.

a: RATIONALE FOR CHOOSING CONVOLUTIONAL LSTM

Convolutional LSTM (ConvLSTM) networks are a modification of both CNN and LSTM integrated models. They incorporate convolutional operations in their LSTM model, which means the model is able to capture spatiotemporal features. Specifically, in the context of audio deepfake detection, ConvLSTMs are able to capture both the frequency and time characteristics of the audio signals. This way, ConvLSTMs have a dual utility for tasks in which both local and long-term context are critical. With ConvLSTMs, the proposed model can leverage detailed feature extraction with CNNs and sequential modeling with LSTMs.

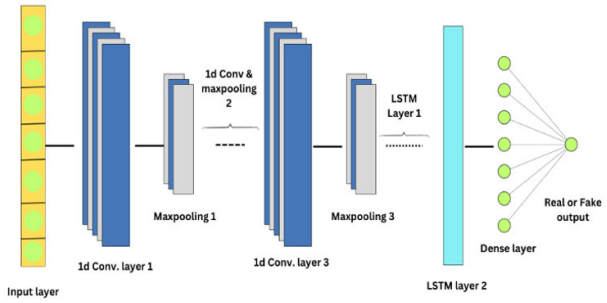

FIGURE 9. ConvLSTM architecture.

TABLE 3. Experimental setup hyperparameters.

Models	Parameter	Value
RNN	Batch Size	32
	Epochs	15
	Activation Function	ReLu
	Optimizer	Adam
	Loss Function	binary_crossentropy
	Regularizer	L2(0.01)
1D CNN	Batch Size	4
	Epochs	10
	Activation Function	ReLu
	Optimizer	Adam
	Loss Function	binary_crossentropy
LSTM	Batch Size	4
	Epochs	10
	Optimizer	Adam
	Loss Function	binary_crossentropy
ConvLSTM	Batch Size	4
	Epochs	10
	Optimizer	Adam
	Loss Function	binary_crossentropy
Proposed Ensemble Model	Batch Size	32
	Epochs	10
	Optimizer	Adam
	Loss Function	binary_crossentropy
	Learning Rate	1e-3
	Stopping Criteria	Early Stopping

E. EXPERIMENTAL SETUP

Colab Pro is used for experimentation with 51 GB of RAM and 12 GB of GPU. Table 3 summarizes all there is to know about the hyperparameters used in experiments across the different machine learning models, with specific configurations used to optimize performance. The models include an RNN (Recurrent Neural Network), a 1-dimensional Convolutional Neural Network (1D CNN), an LSTM (Long Short-Term Memory) network, a ConvLSTM (Convolutional LSTM), and the proposed ensemble model.

1) PARAMETER TUNING RATIONALE

Tuning parameters is an important part of training machine learning models because it affects how well they work and their ability to generalize to other situations. This is why it is necessary to tune every ECN-MF parameter. A common feature among many of these models is the use of similar hyperparameters, including batch size, number of epochs, activation function (ReLU), optimizer (Adam), and loss function (binary_crossentropy). The Adam optimizer was set with a learning rate of 0.001 to provide a balanced approach without overshooting optimal parameters. To prevent overfitting, an L2 regularization term with a coefficient of 0.01 was utilized. This helps control the model's complexity and makes it better at fitting new data sets. Both the 1D CNN and ConvLSTM models utilize a smaller batch size of 4, allowing the model to generalize better by training with fewer samples per batch. By updating the model's weights more frequently and making it more responsive to changes, smaller batch sizes may help accelerate learning from data. The proposed ensemble model, designed to combine the strengths of different model architectures, features additional parameters such as learning rates of 0.001 and mechanisms to reset the network when it starts overtraining. Additionally, it serves as a platform for fine-tuning each model from a training standpoint, adopting an approach that targets precision and effectiveness in tasks involving binary classification.

2) POTENTIAL SOURCES OF VARIABILITY

To ensure the results of the experiment are solid and reliable, it is important to identify and control possible sources of variation:

- **Dataset Variability:** The variety and representativeness of the training data greatly affect the model's performance. To address this, the Fake-or-Real dataset and its sub-datasets were used to analyze a wide range of audio features and preprocessing techniques.
- **Random Initialization:** Initial weight values play an integral part in neural network models, and running the model multiple times with different random seeds can indicate its overall stability and consistency in performance.
- **Hardware Differences:** Changes in computing resources, such as GPU and RAM, can affect training times and results. Ensuring consistent hardware setup or documenting the hardware specifications can help replicate results.

F. EVALUATION MEASURES

After completing the deep learning models and the proposed ensemble model, the primary objective was to assess their performance, specifically regarding model predictions. While accuracy is a commonly used metric to gauge a model's learning effectiveness, it alone may not be sufficient. Therefore, additional metrics such as recall, precision, and F1 score are often calculated to better understand the model's performance. The F1 score, the harmonic mean of recall and precision, ranges from zero to one, with higher values indicating better performance. We used all the above metrics for the deep learning models to evaluate the proposed ensemble model. The details are as follows:

- **True Positives (TP):** These represent the positive instances correctly classified as positive.
- **True Negatives (TN):** This category encompasses the negative instances accurately identified as negative.

- **False Positives (FP):** This category accounts for the negative instances mistakenly classified as positive.
- **False Negatives (FN):** These include the positive instances that were erroneously classified as negative.

Additionally, we conducted an evaluation using various performance metrics, including Accuracy, Precision, Recall, and F1 Score, alongside the previously mentioned confusion matrix and ROC curve analysis. The details of the performance metrics are as follows:

1) ACCURACY

Accuracy provides an overall measure of our model's ability to correctly classify instances, giving us an essential gauge of its general predictive accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

2) PRECISION

Precision enables us to assess the model's ability to accurately identify positive instances, minimize false positive errors, and ensure the quality of positive predictions. This metric is particularly important in scenarios where the cost of a false positive is high.

$$Precision = \frac{TP}{(TP + FP)} \tag{9}$$

3) RECALL

Recall, also known as sensitivity, allows us to evaluate our model's capability to correctly identify all positive instances, minimize false negatives, and ensure comprehensive coverage of actual positives.

$$Recall = \frac{TP}{(TP + FN)} \tag{10}$$

4) F1 SCORE

The F1 Score offers a balanced evaluation by combining precision and recall, providing a single value representing the model's overall performance.

$$F1Score = 2 \times \frac{Precision \times Recall}{(Precision + Recall)} \tag{11}$$

The evaluation process included using a wide range of metrics, justified by the confusion matrix, and involved a thorough ROC (Receiver Operating Characteristic) curve analysis [34]. This comprehensive and detailed model evaluation process provided a holistic approach to understanding the models' impact and revealed their complex performance in a multi-dimensional arena. We assessed the model from an accuracy standpoint by allowing a sharp distinction between positives and negatives and by examining its bias as the decision threshold changes. The results are bound to give us a holistic view. The evaluation tools not only facilitated testing of the model but also made us aware of its vulnerabilities and the aspects that influence its ability to handle the complexity of real-world problems. In conclusion, this precise assessment framework contributed not only to the numerical data but also to the practical meaning of the results, which played a significant part in comprehending the model's durability and potential applications.

IV. RESULTS AND DISCUSSION

Careful analysis of results across various datasets allows us to draw interesting conclusions about the model's generalizability and its reliable performance in practical settings. Four separate datasets form the basis of the experimental results:

1) Performance evaluation of the proposed technique on the "for-orig" dataset.
2) The "for-norm" dataset is employed for evaluating the proposed method.
3) The "for-rerec" dataset is used to test the suggested strategy.
4) The "for-2sec" dataset is used to test the proposed method.
5) Comparative analysis of the proposed method with state-of-the-art methods.

A. PERFORMANCE ANALYSIS OF THE PROPOSED METHOD ON THE FOR-ORIG DATASET

Table 4 displays the performance metrics of different machine learning models on the for-orig dataset. It demonstrates their efficacy using five essential metrics: Accuracy (%), Precision, Recall, F1 Score, and Area Under the Curve (AUC). The evaluated models include Recurrent Neural Networks (RNN), 1D Convolutional Neural Networks (1D CNN), Long Short-Term Memory Networks (LSTM), Convolutional LSTM (ConvLSTM), and an Ensemble Approach. The Ensemble Approach exhibits outstanding performance, achieving perfect scores in Precision, Recall, F1 Score, and AUC, with an accuracy rate of 99.50%. The 1D CNN model closely aligns with a nearly flawless accuracy rate of 99.60% and excels in all other metrics. The LSTM and ConvLSTM models demonstrate exceptional performance, particularly in AUC, with all models achieving a perfect score of 100%. Although the RNN is the least accurate, it still achieves an excellent accuracy rate of 98.80%. Table 4 demonstrates the notable effectiveness of various models in the given task, with the Ensemble Approach slightly surpassing the others.

TABLE 4. Testing results of the proposed model on the for-orig dataset.

Models	ACC	Precision	Recall	F1 Score	AUC
RNN	98.80%	98%	100%	99%	100%
1D CNN	99.60%	99%	100%	100%	100%
LSTM	99.30%	99%	100%	99%	100%
ConvLSTM	98.50%	98%	99%	99%	100%
Ensemble Model	**99.50%**	**100%**	**100%**	**100%**	**100%**

When applied to the for-orig test set, the proposed ensemble model's performance is shown by the confusion

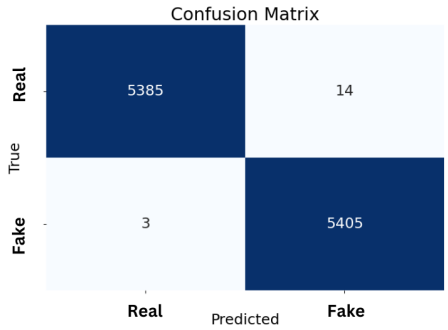

FIGURE 10. The proposed ensemble model confusion matrix on the for-orig test set.

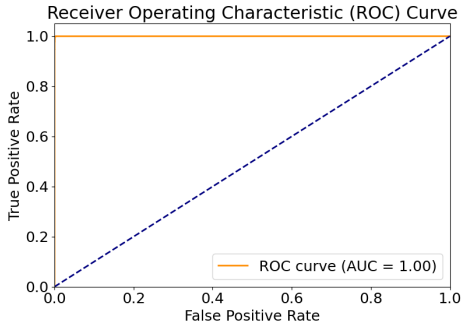

FIGURE 11. The proposed ensemble model ROC curve on the for-orig test set.

matrix in Figure 10. The predicted labels are divided into four quadrants in the classification results matrix by the actual labels. With 5,385 'Real' cases accurately detected by the model, the top-left quadrant represents the True Positives (TP). In the bottom-right quadrant, 5,405 'Fake' instances are accurately recognized, representing the True Negatives (TN). There are fewer misclassifications by the model. Specifically, 14 'Real' instances were mistakenly labeled as 'Fake' in the top-right quadrant, representing False Positives (FP). In the bottom-left quadrant, 3 'Fake' instances were incorrectly classified as 'Real', representing False Negatives (FN). The ensemble model has shown impressive predictive accuracy, with high True Positive and True Negative rates, indicating that it effectively differentiates between 'Real' and 'Fake' classifications in the dataset. With such a small number of incorrect predictions, the model is robust and reliable.

When applied to the for-orig test set, the suggested ensemble model yielded the ROC curve shown in Figure 11. The receiver operating characteristic (ROC) curve plots the performance of a classifier against a range of threshold values. In this graph, the y-axis represents the True Positive Rate (TPR), while the x-axis represents the False Positive Rate (FPR). As the classifier's decision threshold changes, the curve follows the trade-off between specificity (1 - FPR) and sensitivity (or TPR). The test's accuracy is proportional to how closely the curve matches the left-hand and top-left borders of the ROC space. The ensemble model's orange ROC curve hugs the plot's upper left corner, showing a low false positive rate and a high true positive rate for almost all thresholds. Overlapping the dashed blue line is the effective model's curve, while the random classifier's baseline (chance) performance is below it. The legend provides the AUC as 1.00, a perfect score. The AUC indicates how well a classifier can distinguish between positive and negative categories. Excellent prediction accuracy is marked by an AUC of 1.00, indicating that the model has a 100% chance of properly differentiating between a random 'Real' instance and a random 'Fake' instance. Figure 11 visually confirms the model's diagnostic ability and highlights its effectiveness.

B. PERFORMANCE OF THE PROPOSED METHOD ON THE FOR-NORM TEST SET

The results of various machine learning models' testing on the for-norm test set are summarized in Table 5. An F1 score of 98%, an accuracy of 98%, a precision of 97%, and an AUC of 99% demonstrated that the suggested ensemble model performed strongly. These metrics reflect how well the model's predictions correspond to the actual classifications, showing high congruence due to its balanced precision and recall, and they do not lead to memorizing incorrect data. A near-flawless AUC score of 99% demonstrates the model's exceptional class distinction capabilities. This level of effectiveness allows the neural network to reliably differentiate between distinct classes and situations, resulting in outstanding overall performance. Furthermore, the model maintains these high percentages, achieving F1 scores in the upper 90s, along with exceptional accuracy and precision. Compared to the proposed ensemble model and the 1D CNN, the LSTM and RNN models are competent but not extraordinary. Their accuracy rates exceed 90%, and they achieve an AUC of 98%. Despite its somewhat lower precision, the ConvLSTM model has been tested against similar models and has shown solid results. However, in terms of recall rate and maintaining a constant area under 98% (AUC), the 1D CNN is the only exception. The effectiveness and comparative superiority of the proposed ensemble model across all evaluated parameters are highlighted in Table 5, indicating its potential for reliable prediction performance in practical applications.

TABLE 5. Testing results of the proposed model on the for-norm test set.

Models	ACC	Precision	Recall	F1 Score	AUC
RNN	93.10%	91%	96%	93%	98%
1D CNN	97.30%	96%	99%	97%	100%
LSTM	94.00%	92%	96%	94%	98%
ConvLSTM	90.40%	85%	99%	91%	98%
Ensemble Model	**98%**	**97%**	**98%**	**98%**	**99%**

As seen in Figure 12, the model's classification capabilities are reflected in the confusion matrix of the proposed ensemble model when tested on the for-norm test set. The

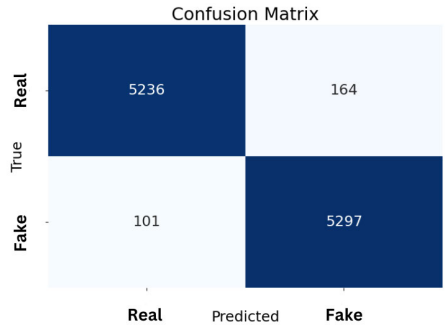

FIGURE 12. The proposed ensemble model confusion matrix on the for-norm test set.

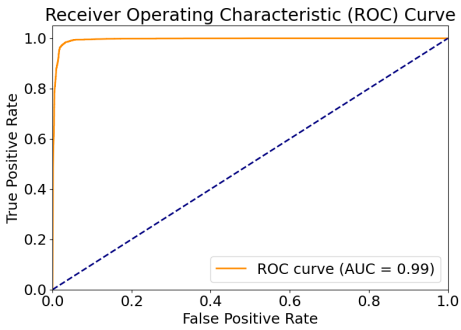

FIGURE 13. The proposed ensemble model ROC curve on the for-norm test set.

model identified 5,236 cases as 'Real' (True Positives) and accurately classified 5,297 instances as 'Fake' (True Negatives), along with the results shown in the matrix. The model did make some mistakes; for example, it erroneously labeled 164 'Real' instances as 'Fake' (False Positives) and 101 'Fake' instances as 'Real' (False Negatives). While the model does a good job overall, there is an opportunity for improvement in lowering the number of cases where the 'Real' and 'Fake' classes are confused, as these misclassifications show. These findings highlight the model's capabilities for predicting actual classes and its shortcomings when distinguishing between them.

The ROC curve for the proposed ensemble model, as tested on the for-norm test set, is shown in Figure 13. According to the legend, the AUC is 0.99, which translates to 99%, which is exceptional. We have perfect classification with an AUC of 99%. With the proposed ensemble model providing great discriminative power between the 'Real' and 'Fake' categories in the test set, an AUC of 0.99 denotes a high level of model performance. The diagonal dashed line represents the ROC curve of a random classifier for comparison, underscoring the superiority of the proposed ensemble model's performance above random chance.

C. PERFORMANCE OF THE PROPOSED METHOD ON THE FOR-2SEC TEST SET

The performance of different machine learning models on the for-2sec test set is compared in detail in Table 6.

TABLE 6. Results of the proposed model on the for-2sec test set.

Models	ACC	Precision	Recall	F1 Score	AUC
RNN	90.30%	88%	93%	91%	96%
1D CNN	95.60%	95%	97%	96%	99%
LSTM	89.90%	90%	90%	90%	96%
ConvLSTM	95.00%	93%	97%	95%	99%
Ensemble Model	**96.90%**	**98%**	**96%**	**97%**	**100%**

Outstanding results were achieved by the proposed ensemble model, which topped the table with a 96.90% accuracy rate, indicating that it correctly identified most occurrences. It was also the most accurate in predicting the correct class instances because of the precision score of 98%. It correctly detected 96% of all the relevant labels since the recall score of the model is 96%. A 97% F1 Score, which is the combination of recall and precision, shows the robustness of the model. The model's remarkable ability to accurately discriminate between the classes was demonstrated by its flawless AUC score of 100%. The 1D CNN and ConvLSTM models exhibited an AUC of 99%, showing both models' strong performance. They performed very well in accuracy, precision, recall, and F1 scores because they all scored in the 90s. Also, considering the other models, RNN and LSTM showed respectable performances with accuracy levels of approximately 90%. The F1 score of 91% for RNN and 90% for LSTM indicates that RNN has a marginally superior precision-recall balance compared to that of the LSTM. When looking at the AUC percentage, it is clear that the proposed ensemble model has an excellent balance of accuracy, precision, recall, and discriminatory power. Based on these metrics for the for-2sec test set categorization assignment, the model is quite effective and dependable.

For the proposed ensemble model tested on the for-2sec test set, the confusion matrix is shown in Figure 14. Showing significant skill in correctly categorizing both groups, the model correctly predicted 1,380 cases as 'Real' (True Positives) and 1,355 instances as 'Fake' (True Negatives). A total of 33 cases were wrongly forecast as 'Fake' when they were actually 'Real' (False Positives), and 58 instances were incorrectly predicted as 'Real' when they were actually 'Fake' (False Negatives), proving that the model was not fail-safe. With more correct classifications than erroneous ones, this confusion matrix shows that the model's predictions are accurate. The model distinguishes between true and made-up data with a high degree of accuracy and reliability, and even for this test set, we only found a few false positives and false negatives. This study's findings demonstrate that the proposed ensemble model can reliably perform classification tasks.

Based on its evaluation of the for-2sec test set, the ROC curve for the proposed ensemble model is displayed in Figure 15. The ROC curve, a graph portraying the TPR-FPR trade-off at various threshold levels, helps illustrate this dynamic. A good classifier will have a high TPR

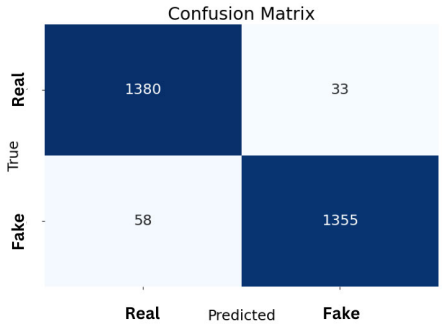

FIGURE 14. The proposed ensemble model confusion matrix on the for-2sec test set.

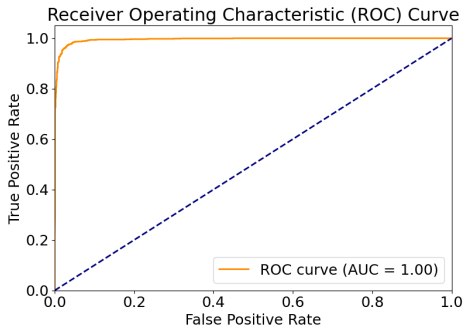

FIGURE 15. The proposed ensemble model ROC curve on the for-2sec test set.

and a low FPR, as shown by the sharply ascending curve towards the upper-left corner. The model's high degree of sophistication is showcased with the AUC score of 1.00, which indicates 100% accuracy for differentiating between the 'Real' and 'Fake' classes. This information is underlined in the legend. An AUC of 1.00 indicates that the model can accurately distinguish between classes with no ambiguity. The dashed diagonal line represents a random classifier lacking discriminative capability; the large difference between this line and the model's ROC curve shows how much better the model makes predictions. The proposed ensemble model's ability to accurately classify the for-2sec test set is demonstrated by its highly precise predictions, as seen in the graph.

D. PERFORMANCE OF THE PROPOSED METHOD ON THE FOR-REREC TEST SET

Table 7 displays the outcomes of the for-rerec test set conducted on different machine learning models. Among the evaluated models, the proposed ensemble model demonstrated the best performance metrics, with a 92.80% accuracy rate, which accurately detected most instances. With a precision score of 94%, it is a solid predictor of positive outcomes. The model correctly caught many relevant cases, as evidenced by its recall rate of 92%. The model's balanced performance is seen in the 93% F1 Score, the harmonic mean of recall and precision. The model also demonstrated impeccable caliber in distinguishing classes with an AUC

TABLE 7. Results of the proposed model on the for-rerec test set.

Models	ACC	Precision	Recall	F1 Score	AUC
RNN	87.40%	87%	89%	88%	94%
1D CNN	91.40%	90%	93%	92%	97%
LSTM	88.50%	89%	88%	88%	95%
ConvLSTM	90.30%	91%	89%	90%	96%
Ensemble Model	**92.80%**	**94%**	**92%**	**93%**	**98%**

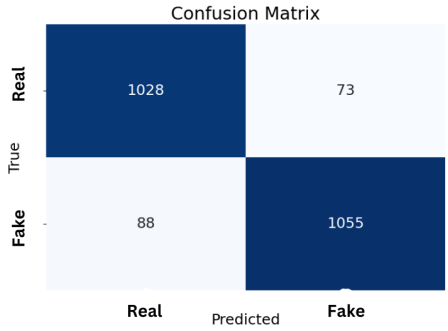

FIGURE 16. The proposed ensemble model confusion matrix on the for-rerec test set.

of 98%. Like the DCNN model, the Conv-LSTM model showed consistent but slightly inferior performance, with an AUC of 96%; however, both models produced solid findings. As can be observed, the RNN and the LSTM models underperformed concerning the overall performance metrics, although they achieved accuracy rates of 87.30% and 88.50%, respectively. Overall, the for-rerec test set demonstrates that the proposed ensemble model outperforms the individual RNN, 1D CNN, LSTM, and Conv-LSTM models regarding accuracy, precision, recall, and AUC, proving its effectiveness and dependability in classification tasks.

For the proposed ensemble model tested on the for-rerec test set, the confusion matrix is shown in Figure 16. Showing a significant skill in correctly categorizing both groups, the model correctly predicted 1,028 cases as 'Real' (True Positives) and 1,055 instances as 'Fake' (True Negatives). 73 cases were wrongly forecast as 'Fake' when they were actually 'Real' (False Positives), and 88 instances were incorrectly predicted as 'Real' when they were actually 'Fake' (False Negatives). With more correct classifications than erroneous ones, this confusion matrix shows that the model's predictions are accurate. The model distinguishes between true and made-up data with a high degree of accuracy and reliability, and even for this test set, we only found a few false positives and false negatives. This study's findings demonstrate that the proposed ensemble model can reliably perform classification tasks.

Based on its evaluation of the for-rerec test set, the ROC curve for the proposed ensemble model is displayed in Figure 17. The model's high degree of sophistication is showcased with the AUC score of 0.98, which indicates 98% accuracy for differentiating between the 'Real' and 'Fake'

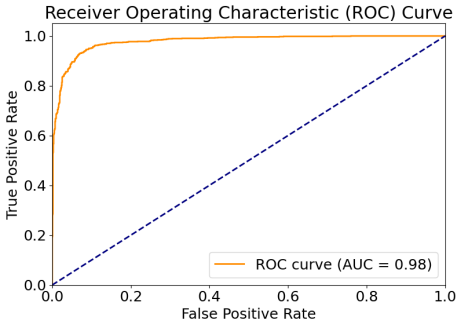

FIGURE 17. The proposed ensemble model confusion matrix on the for-rerec test set.

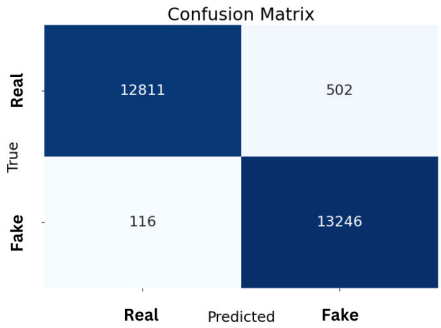

FIGURE 18. The proposed ensemble model confusion matrix on the for-merged test set.

classes. This information is underlined in the legend. An AUC of 0.98 indicates that the model can accurately distinguish between classes with little to no ambiguity. The proposed ensemble model's ability to accurately classify the for-rerec test set is demonstrated by its highly precise predictions, as seen in the graph.

E. PERFORMANCE OF THE PROPOSED METHOD ON THE FOR-MERGED TEST SET

Table 8 shows how different ML models fared when tested on the combined dataset. With an impressive 98% accuracy, the proposed ensemble model significantly surpasses the other models, suggesting a very high percentage of correct predictions. Most of the model's positive predictions are correct, as its precision rate is 96%. The model's recall is 99%, correctly identifying nearly all real positive events. A very balanced performance between precision and recall is reflected in the 98% F1 Score, which is the harmonic mean of the two measures. The model's remarkable ability to distinguish between the classes without overlap is shown by its perfect AUC score of 100%. With equal F1 scores of 96% and an AUC of 99%, the 1D CNN and ConvLSTM models demonstrate outstanding performance, although their accuracies are 96%. The LSTM model manages a respectable AUC of 99%, but its accuracy is 94%, putting it marginally below 1D CNN and ConvLSTM. The RNN model manages a high recall of 99% and an F1 Score of 92%. Compared to LSTM, 1D CNN, and ConvLSTM models, its 99% AUC is comparable, but the accuracy of RNN is 91%, which is low compared to all other models. The for-merged test set shows that the proposed ensemble model performs exceptionally well on all measures, indicating that it is a solid choice for classification tasks with this data type.

By analyzing the results of the for-merged test set, Figure 18 displays the confusion matrix for the proposed ensemble model. With 12,811 'Real' instances classified as 'Real' (True Positives) and 13,246 'Fake' instances classified as 'Fake' (True Negatives), the model correctly recognized a considerable proportion of occurrences, according to the matrix. Nevertheless, there were a few instances of misclassifications in the model. Specifically, it mistakenly forecasted 502 instances as 'Fake' when, in fact, they were 'Real' (False Positives), and it misclassified 116 instances as 'Real' when, in fact, they were 'Fake' (False Negatives). With a higher proportion of accurate classifications in the for-merged test set and a lower proportion of false negatives and false positives, the proposed ensemble model performs well. Although the recall of the model is high and the number of false negatives is low, the former means that the model can easily identify the 'Real' instances. At the same time, the latter shows that the model is good at detecting false negative instances. This confusion matrix verifies the model's high capacity to distinguish between 'Real' and 'Fake' classifications in this dataset.

Figure 19 shows the ROC curve for the proposed ensemble model tested on the for-merged test set. According to the legend, this curve stands out due to its AUC score of 1.00. A model with an AUC of 1.00 can reliably distinguish

TABLE 8. Results of the proposed model on the for-merged test set.

Models	ACC	Precision	Recall	F1 Score	AUC
RNN	91.00%	86%	99%	92%	99%
1D CNN	96.00%	93%	99%	96%	99%
LSTM	94.00%	93%	96%	94%	99%
ConvLSTM	96.00%	94%	98%	96%	99%
Ensemble Model	**98.00%**	**96%**	**99%**	**98%**	**100%**

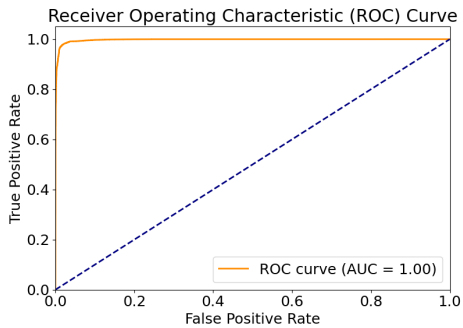

FIGURE 19. The proposed ensemble model confusion matrix on the for-merged test set.

TABLE 9. Comparative analysis of the proposed method with latest state-of-the-art techniques.

Ref.	Features	Models	Dataset	Accuracy (%)
[9]	MFCC-40, Roll-off point, Centroid, Contrast, bandwidth	LSTM VGG16	for-original	91 93
[35]	MFCC-20	SVM RF XGB KNN	for-original	67 62 59 62
[32]	Timber model analysis (Brightness, Hardness, Depth, Roughness)	NB SVM DT(J48) RF	for-norm	67.27 73.46 70.26 71.47
	STFT, Mel-Spectrograms, MFCC, CQT	VGG16 VGG19	for-2sec	92 92
	Waveform classification	VGG19	for-2sec	89.79
Proposed Method	**MFCC-13, Chroma-based, ZCR (Zero Crossing Rate)**	**Ensemble Model**	**for-original for-norm for-2sec for-rerec for-merged**	**99.5 98 96.9 92.8 98**

between the 'Real' and 'Fake' classes. A model with such discriminative performance must have a remarkable knack for picking up on all the subtleties in the dataset. As the dotted diagonal line shows, a no-skill classifier with an AUC of 0.5 would randomly predict the class. The proposed ensemble model's exceptional classification ability on the for-merged test set is further demonstrated by the large departure of its ROC curve from this no-discrimination line.

In conclusion, most test sets, including for-orig, for-norm, for-2sec, for-rerec, and for-merged, have shown that the proposed ensemble model consistently outperforms state-of-the-art models in identifying real and fake occurrences. Compared to other models like RNN, 1D CNN, LSTM, and ConvLSTM, this model consistently outperforms them across all datasets in terms of accuracy, precision, recall, F1 scores, and AUC scores. The model's high True Positive and True Negative numbers in its confusion matrices across all test sets demonstrate its excellent case classification performance. The model's accuracy and dependability were highlighted by the relatively small numbers of False Positives and False Negatives, though there were some. AUC scores are close to or even at 100%. The ROC curves thus indicate an excellent model differentiation between various classes, as shown. Its flawless AUC of 1.00 for the for-merged test set demonstrated the model's exceptional predictive potential. Therefore, the results prove that the proposed ensemble model is a reliable classifier that can always obtain perfect results even when the dataset or environment remains unchanged. Tasks requiring complex separation between categories like 'Real' and 'Fake' are well-suited to its capacity to balance sensitivity and specificity, contributing to its precise predictions.

F. COMPARATIVE ANALYSIS OF THE PROPOSED METHOD WITH STATE-OF-THE-ART METHODS

This study evaluates the effectiveness of our proposed model by comparing its accuracy with that of baseline studies. The evaluation process is carried out under similar experimental conditions, including the dataset and the data samples employed in the source study. Notably, the dataset employed in our investigation is relatively new and has seen limited use in prior studies. In Table 9, we can see how the proposed method stacks up against the most cutting-edge methods for audio classification. Various datasets with different feature sets and models demonstrate the evaluation.

The study [9] uses MFCC-40 and other audio descriptors with LSTM and VGG16 models for the for-original dataset, achieving accuracies of 91% and 93%, respectively. To contrast this, researchers [35] use the same dataset to evaluate more basic models such as SVM, RF, XGB, and KNN using an MFCC-20 feature set; the results show far lower accuracies, ranging from 59% to 67%. In research [32], a model for wood analysis is presented that makes use of the for-norm dataset with classifiers like NB, SVM, DT (J48), and RF; the resulting accuracies range from 67.27% to 73.46%. Using the for-2sec dataset and VGG16 and VGG19 models, the same reference achieves 92% accuracy by utilizing STFT, Mel-Spectrograms, MFCC, and CQT features. Also, using VGG19 for waveform classification on for-2sec results in an accuracy of 89.79%.

The proposed method exhibited excellent outcomes, employing a broad collection of features, including MFCC-13, Chroma-based features, and Zero Crossing Rate (ZCR). On the for-original dataset, the ensemble model exhibited an astounding 99.5% accuracy. This high level of performance was continued across additional datasets as well, with 98% on the for-norm, 96.9% on the for-2sec, 92.8% on the for-rerec, and an amazing 98% on the for-merged dataset (Table 9).

The results show that the proposed method is the best model for audio classification tasks and consistently outperforms other models on different datasets. It also proves robust and effective, setting a new standard for the field.

G. MITIGATION OF OVERFITTING

Overfitting [36] occurs when a model learns the noise and specific patterns of the training data to a level where the model performs very well on the training examples but poorly

on new/test data. To mitigate overfitting, several techniques were employed during the model development process.

1) REGULARIZATION

L2 regularization, also known as weight decay, was used to reduce the weights to small values, and thus, the model aims to generalize well on other datasets beyond those it was trained on.

2) DROPOUT

Dropout layers were included in the neural network architectures, in which some units were randomly dropped during training to reduce the model's dependence on certain neurons and to enhance its generalization capability.

3) EARLY STOPPING

During training, early stopping was utilized to monitor the model's performance on a validation set. Training was stopped once the performance on the validation set no longer improved, thus reducing the model's tendency to overfit the training data.

4) Kfold CROSS VALIDATION

In order to get better estimations and reduce the risk of overfitting, the evaluation was carried out on four subsets of the dataset through 3-fold cross-validation. In each of the subsets, there was a training, testing, and validation directory. The data and the labels were then joined together, and 3-fold cross-validation was done on them.

The overall proposed architecture, containing RNN, 1D CNN, LSTM, and ConvLSTM, also opted for 3-fold cross-validation. Table 10 provides the average performance of each of the base models and the ensemble model based on the results of the validations.

Thus, in 3-fold cross-validation, the model is accurate and capable of generalizing the dataset as it achieves high accuracy. The fact that the model demonstrated such good performance on diverse datasets combined with different audio conditions shows that the model was learning the right patterns and features in the datasets for deepfake detection and did not overfit the features of the data. With L2 regularization, dropout layers, early stopping, and cross-validation, the exploring model alleviates overfitting and improves the model's performance. These measures help to verify that the achieved high accuracies do not impose an overfitting effect due to multiple iterations through a specific issue but are rather true to the model's aptitude in detecting audio deepfakes under varying conditions.

TABLE 10. Average 3-fold cross-validation results of all sub datasets.

Datasets	Models	ACC	Prec.	Rec.	F1 Score	AUC
for-orig	RNN	98%	99%	98%	98%	100%
	1D CNN	99.6%	100%	100%	100%	100%
	LSTM	99%	99%	100%	100%	100%
	ConvLSTM	99%	100%	99%	99%	100%
	Ensemble Model	**99.7%**	**100%**	**100%**	**100%**	**100%**
for-norm	RNN	95%	95%	96%	97%	99%
	1D CNN	98%	98%	98%	98%	100%
	LSTM	97%	97%	98%	98%	100%
	ConvLSTM	98%	98%	99%	99%	100%
	Ensemble Model	**99%**	**99%**	**99%**	**99%**	**100%**
for-2sec	RNN	75%	71%	86%	78%	81%
	1D CNN	96%	96%	96%	96%	99%
	LSTM	93%	89%	97%	93%	98%
	ConvLSTM	95%	97%	98%	98%	100%
	Ensemble Model	**99%**	**98%**	**99%**	**99%**	**100%**
for-rerec	RNN	94%	93%	94%	94%	99%
	1D CNN	97%	97%	97%	97%	100%
	LSTM	97%	97%	97%	97%	100%
	ConvLSTM	98%	98%	99%	99%	100%
	Ensemble Model	**99%**	**99%**	**99%**	**99%**	**100%**

H. THEORETICAL ANALYSIS OF THE EFFECTIVENESS OF THE ENSEMBLE MODEL

Considering the results presented in Table 8 over the "for merged" test set, this indicates that the ensemble models performed better than the individual models. These results are further analyzed through theoretical concepts, including bias-variance trade-off, error rate bounds, and convergence properties, as follows:

1) BIAS-VARIANCE TRADE-OFF

1) **Reduction of Variance:** It is observed that the ensemble approach effectively reduces the variance component of prediction errors. Variance refers to the model's sensitivity to fluctuations in the training data. By combining multiple models, each trained with different data, the ensemble approach averages out the prediction errors. This reduction in variance leads to more stable and robust performance. The accuracies of individual models range from 91% to 96%, wheras the ensemble achieves a higher accuracy of 98%, indicating a significant reduction in variance.
2) **Mitigation of Bias:** Individual models may have specific biases due to their structure and training process. The ensemble method mitigates these biases by combining various models trained with different perspectives and parameters. For example, an RNN model captures sequential dependencies well, while a CNN model effectively captures local patterns. The ensemble of diverse models can mitigate individual biases, leading to better accuracy and generalization.

2) ERROR RATE BOUNDS

1) **Error Rate Reduction:** Table 8 demonstrates that the ensemble model's error rate is theoretically lower than the average error rates of the individual models. If the errors of individual models are uncorrelated, the ensemble error rate ϵ_e is bounded above the average of individual error rates. Considering the accuracies of the individual models, the ensemble model's accuracy

of 98% indicates a lower combined error rate, offering the advantage of error diversification.

$$\epsilon_e \leq \frac{1}{N}\sum_{i=1}^{n}\epsilon_i \tag{12}$$

where n is the number of models. Given individual accuracies of 91%, 96%, 94%, and 96%, the ensemble achieves a higher accuracy than the average of the individual errors.

2) **Improved Precision and Recall:** The precision and recall values presented in Table 8 demonstrate that the ensemble model achieves a precision of 96% and a recall of 99%, both higher than those of the individual models. This indicates the superiority of the ensemble model over the individual models in correctly identifying true positives (high recall) and reducing false positives (high precision). The F1 measure of 98% further illustrates the balance and robustness of the ensemble approach, significantly improving the reliability of the predictions.

3) CONVERGENCE PROPERTIES

1) **Faster and More Stable Convergence:** Ensemble methods tend to converge faster and more stably than individual models. The combined learning process benefits from the strengths of each base model, leading to more consistent learning outcomes. This explains why the ensemble model achieved a near-perfect AUC (100%) and a high F1 measure (98%). These results indicate the model's proficiency in distinguishing between classes and maintaining performance across different metrics effectively.
2) **Generalization Across Datasets:** The performance measures indicate that the ensemble approach generalizes better across different datasets and conditions. Each base model captures different aspects of the data, and their combination ensures the ensemble model's robustness against varied inputs. This is reflected in the high AUC of 100%, indicating the ensemble's excellent generalization capability and robustness.

The theoretical analysis concepts indicate that the ensemble model's effectiveness stems from its ability to balance and reduce bias and variance, lower overall error rates through model diversity, and ensure faster and more stable convergence. By leveraging the complementary strengths of RNN, 1D CNN, LSTM, and ConvLSTM, the ensemble model achieves superior performance metrics, such as higher accuracy, precision, recall, F1 measure, and AUC. These theoretical insights, supported by empirical results, confirm the ensemble approach's robustness and efficacy in complex predictive tasks.

I. POTENTIAL CHALLENGES OR LIMITATIONS

The proposed ECN-MF model is highly effective in deepfake detection but encounters several challenges and limitations, particularly related to computational complexity and scalability. One primary concern is the high computational demand to train and deploy the ensemble model. ECN-MF combines multiple complex base models (RNN, 1D CNN, LSTM, and ConvLSTM), and each base model requires substantial computational resources. The process of training these models concurrently is resource-intensive and requires a high-performance computing infrastructure that may not be accessible in all scenarios. Additionally, the training phase is time-consuming; each base model must be trained individually before combining their predictions, leading to a longer training time than a single model. This extended training period may become a bottleneck, especially in environments where rapid model updates are essential. Moreover, the ensemble method demands significant memory resources to manage the parameters and intermediate computations of multiple models, which poses a challenge in contexts with limited memory capacity, such as embedded systems.

Scalability presents another set of issues. While ECN-MF performs well on the provided datasets, scaling the approach to larger datasets might prove problematic. The computational and memory requirements increase with the dataset size, potentially leading to inefficiencies and slow processing times. Implementing distributed training to handle large datasets and complex models is also complicated, as it involves efficient communication and synchronization across multiple nodes in a distributed environment. That requires additional overhead and potential bottlenecks. Furthermore, the inference phase of the ensemble model involves executing multiple models and combining their results, which results in higher latency. This increased latency might be unacceptable in real-time applications where swift responses are crucial. Additionally, managing and maintaining an ensemble of multiple models adds another layer of complexity. The increased system complexity can make maintenance more challenging, as issues may arise from the interactions between different models. The brief information about the challenges encountered and potential limitations is as follows:

1) CHALLENGES ENCOUNTERED

1) **Variability in Audio Quality:** One of the main difficulties was the inconsistency in the audio quality of the datasets collected. Audio samples differed in length, bit rate, and the level of background noise, necessitating preprocessing procedures to transform the samples into a more suitable form.
2) **Computational Complexity:** A large amount of time and computational resources were required to train the ensemble model with the architectures of RNN, 1D-CNN, LSTM, and ConvLSTM. The complexity of the current model led to longer training times and required the use of a high-end processor.

2) POTENTIAL LIMITATIONS

1) **Generalizability to Unseen Data:** Although testing was conducted on testing subsets, the model's ability to generalize to new and diverse datasets remains a limitation. It should be noted that the presented approach has not been extensively tested on a wide variety of datasets, so more testing is needed to fully ascertain its stability.
2) **Overfitting Risk:** While techniques such as regularization, dropout, and early stopping were employed, the complexity of the model still poses certain overfitting risks. This issue can be addressed through ongoing monitoring and further validation to prevent overfitting.
3) **Real-Time Application:** Some components of the architecture increase computational complexity, which may be disadvantageous in real-time applications. Future work could focus on improving the model to make faster predictions without compromising accuracy.

Overcoming these challenges with accurate preprocessing, appropriate train-test splits, and multiple validation steps will ensure the efficiency and reliability of the proposed approach in identifying audio deepfakes. Further development should aim to increase the model's flexibility and speed to broaden its applications in real-world scenarios.

V. CONCLUSION AND FUTURE WORK

This study utilizes features like MFCCs, Chroma, and Zero Crossing Rate to identify fake audio through deep learning and ensemble learning approaches. The dataset used for this task is the Fake-or-Real dataset, which is divided into four sub-datasets ('for-norm', 'for-original', 'for-2sec', and 'for-rerec') based on bit rate and audio duration. The experiment combined all the sub-datasets from the Fake-or-Real dataset, and four basic models (RNN, 1D CNN, LSTM, and ConvLSTM) were integrated to create an ensemble model. On the 'for-original' sub-dataset, the proposed ensemble model demonstrated a state-of-the-art accuracy of 99.5%, closely matched by the accuracy of the 1D CNN model at 99.6%. Additionally, it achieved 98% accuracy on the 'for-norm' sub-dataset, 96.9% on the 'for-2sec', and 92.8% on the 'for-rerec'. Similarly, the proposed ensemble model achieved an accuracy of 98% using the 'for-merged' dataset, which was constructed by combining all the sub-datasets. Waveform and spectrogram samples derived from real and fake audio demonstrate clear differences, supporting our proposed ensemble model's high accuracy and AUC scores. These differences confirm its ability to classify audio signals effectively. The confusion matrices provide information on how well a model performs by showing its number of true positives, true negatives, false positives, and false negatives. Examining misclassified samples may provide insight into common patterns that cause errors, leading to additional improvements or more training data for future models. The results demonstrate that the proposed method is superior to the individual models, significantly enhancing media content authenticity verification. By consistently outperforming the individual models across all four datasets, the experimental findings highlight the effectiveness of our approach and its applicability to diverse datasets. The suggested method not only increases the likelihood of accurate results but also helps readers understand the deepfake identification process in greater detail.

Future work will explore ensemble learning techniques such as hard voting, stacking, and gradient boosting. Additionally, the investigation will include varying window sizes for MFCC and other critical audio features. While this study focuses on detecting fake audio content, the proposed approach demonstrates versatility by extending its application to identifying fake content in videos and images. This adaptability enhances the methodology's broader applicability across different media types, reinforcing its potential impact on content authenticity verification.

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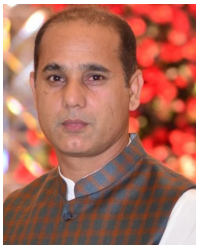

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